

XRP Ledger Node — Windows VPS One-Click Setup

This setup installs and configures an XRP Ledger (rippled) node on a Windows Server 2022 VPS using WSL2 + Ubuntu.

Targets:

- Windows Server 2022
- 4 vCPU+
- 8 GB RAM+
- NVMe SSD

Recommended providers:

- Vultr
- Hetzner
- Kamatera

Step 1 — Create Your VPS

Recommended spec:

Resource	Recommended
OS	Windows Server 2022
CPU	4 vCPU
RAM	8 GB
Disk	160–250 GB NVMe
Network	1 Gbps

After deployment: 1. Connect using Remote Desktop 2. Open PowerShell as Administrator 3. Continue below

Step 2 — One-Click PowerShell Installer

Save this as:

```
install-xrpl-node.ps1
```

Then run:

```
Set-ExecutionPolicy Bypass -Scope Process -Force
.\install-xrpl-node.ps1
```

```
# =====
# XRP Ledger Node Auto Installer
# Windows Server 2022 + WSL2 + Ubuntu
# =====

Write-Host "=====
Write-Host " XRP Ledger Node Installer"
Write-Host "=====

# -----
# Enable WSL + Virtual Machine Platform
# -----

Write-Host "Enabling WSL features..."

dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-
Linux /all /norestart

dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /
norestart

# -----
# Install WSL2
# -----

Write-Host "Installing WSL2..."

wsl --install -d Ubuntu

Write-Host ""
Write-Host "IMPORTANT:"
Write-Host "Reboot the server after this script finishes."
Write-Host "Then run the SECOND STAGE script."
Write-Host ""

pause
```

Step 3 — Reboot VPS

After reboot:

1. Open Ubuntu from the Start Menu
2. Create Linux username/password
3. Continue below

Step 4 — Second Stage Installer

Save this as:

```
install-xrpl-stage2.ps1
```

Run as Administrator:

```
Set-ExecutionPolicy Bypass -Scope Process -Force  
.\install-xrpl-stage2.ps1
```

```
# =====  
# XRP Ledger Node Stage 2 Installer  
# =====  
  
Write-Host "Installing XRPL node inside Ubuntu..."  
  
wsl bash -c "sudo apt update"  
  
wsl bash -c "sudo apt install -y wget curl gnupg2 ufw"  
  
wsl bash -c  
"wget -q -O - https://repos.ripple.com/repos/api/gpg/key/public | sudo gpg --  
dearmor -o /usr/share/keyrings/ripple.gpg"  
  
wsl bash -c "echo 'deb [signed-by=/usr/share/keyrings/ripple.gpg] https://  
repos.ripple.com/repos/rippled-deb bookworm stable' | sudo tee /etc/apt/  
sources.list.d/ripple.list"  
  
wsl bash -c "sudo apt update"  
  
wsl bash -c "sudo apt install -y rippled"  
  
# -----  
# Backup config
```

```

# -----

wsl bash -c "sudo cp /etc/opt/ripple/rippled.cfg /etc/opt/ripple/
rippled.cfg.bak"

# -----
# Enable pruning config
# -----

$config = @"
[node_db]
type=NuDB
path=/var/lib/rippled/db/nudb
open_files=2000
filter_bits=12
cache_mb=1024
online_delete=2000
advisory_delete=0

[database_path]
/var/lib/rippled/db

[ledger_history]
2000

[server]
port_rpc_admin_local
port_peer
port_ws_admin_local

[port_rpc_admin_local]
port=5005
ip=127.0.0.1
admin=127.0.0.1
protocol=http

[port_ws_admin_local]
port=6006
ip=127.0.0.1
admin=127.0.0.1
protocol=ws

[port_peer]
port=51235
ip=0.0.0.0
protocol=peer

[sntp_servers]
time.windows.com
pool.ntp.org
"@

```

```

$tempFile = "$env:TEMP\\rippled.cfg"
$config | Out-File -Encoding ascii $tempFile

wsl bash -c "sudo mkdir -p /etc/opt/ripple"

wsl bash -c "sudo rm -f /etc/opt/ripple/rippled.cfg"

wsl bash -c "sudo cp /mnt/c/Windows/Temp/rippled.cfg /etc/opt/ripple/
rippled.cfg"

# -----
# Open firewall ports
# -----

New-NetFirewallRule -DisplayName "XRPL Peer Port" `
-Direction Inbound `
-Protocol TCP `
-LocalPort 51235 `
-Action Allow

New-NetFirewallRule -DisplayName "XRPL RPC Local" `
-Direction Inbound `
-Protocol TCP `
-LocalPort 5005 `
-Action Allow

New-NetFirewallRule -DisplayName "XRPL WebSocket" `
-Direction Inbound `
-Protocol TCP `
-LocalPort 6006 `
-Action Allow

# -----
# Start rippled
# -----

wsl bash -c "sudo systemctl enable rippled"

wsl bash -c "sudo systemctl restart rippled"

Write-Host ""
Write-Host "====="
Write-Host " XRPL Ledger node installed successfully"
Write-Host "====="
Write-Host ""
Write-Host "Check status with:"
Write-Host "wsl bash -c 'sudo systemctl status rippled'"
Write-Host ""
Write-Host "Check sync with:"

```

```
Write-Host "curl http://127.0.0.1:5005"  
Write-Host ""
```

Step 5 — Verify Sync

Run:

```
wsl bash -c "curl -X POST http://127.0.0.1:5005/ -H 'Content-Type:  
application/json' -d '{\"method\":\"server_info\", \"params\":[]}'"
```

You want to eventually see:

```
"server_state": "full"
```

Monitoring Commands

Service Status

```
wsl bash -c "sudo systemctl status rippled"
```

Live Logs

```
wsl bash -c "journalctl -u rippled -f"
```

Peer Count

```
wsl bash -c "rippled peers"
```

Optional — Auto Restart Watchdog

Save as:

```
xrpl-watchdog.ps1
```

```
while ($true) {  
  
    $status = wsl bash -c "systemctl is-active rippled"  
  
    if ($status -notmatch "active") {  
  
        Write-Host "Rippled stopped. Restarting..."  
  
        wsl bash -c "sudo systemctl restart rippled"  
    }  
  
    Start-Sleep -Seconds 60  
}
```

You can schedule this with Windows Task Scheduler.

Recommended Security

Keep RPC Local Only

Do NOT expose:

- 5005
- 6006

to the public internet unless protected.

Only expose:

51235

for XRPL peer networking.

Optional Next Upgrades

Later you can add:

- Docker deployment
- HTTPS reverse proxy
- NGINX
- Public API endpoint
- Validator mode
- Full-history mode
- Grafana monitoring

- Auto-backups
 - Snapshot restores
 - Cloudflare protection
-

Useful References

Official XRPL documentation:

- <https://xrpl.org/docs.html>

Official rippled repository:

- <https://github.com/XRPLF/rippled>